Lecture-24
Internal Flows III: Turbulent velocity profiles in a fully developed pipe flow

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Learning Objectives

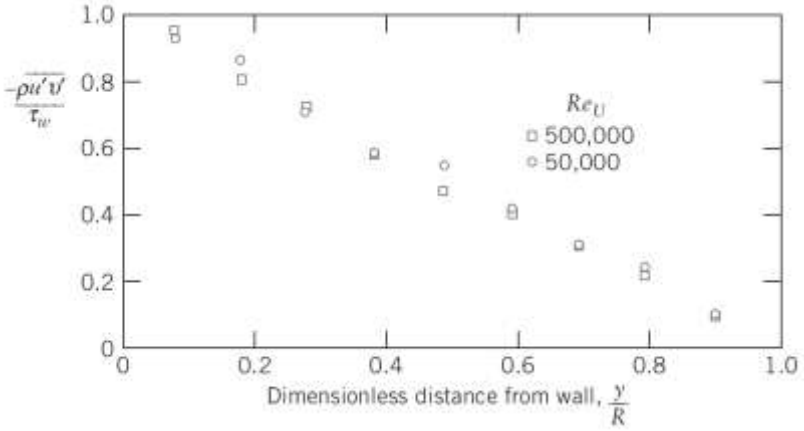
By the end of this lecture, you should be able to

1. Prove that the **total shear stress varies linearly** with radial position in a pipe using momentum balance
2. Describe the spatial shift from **viscous-dominated** stress near the walls to **Reynolds-stress dominance** in the core.
3. Derive the **Reynolds stress** expression using **Prandtl's mixing length hypothesis** and fluctuating velocity scales.
4. Deduce the **linear, viscous sublayer** and **logarithmic** regime.

Shear stress distribution in a fully developed flow

- Both for laminar and turbulent **fully developed flow,** shear stress **varies linearly** across the radius of a pipe.
- The **stress relations for turbulent flow** are primarily based on experimental observations and **correlations**.
- Turbulent flow at each point is represented by a **time-averaged mean component** and **fluctuating components**.
- The fluctuating components **transfer momentum between adjacent fluid layers**. This shows up as an effective stress, called the **Reynolds stress**.

$$\tau = \tau_{\text{lam}} + \tau_{\text{turb}} = \mu \frac{d\bar{u}}{dy} - \rho \overline{u'v'}$$



Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

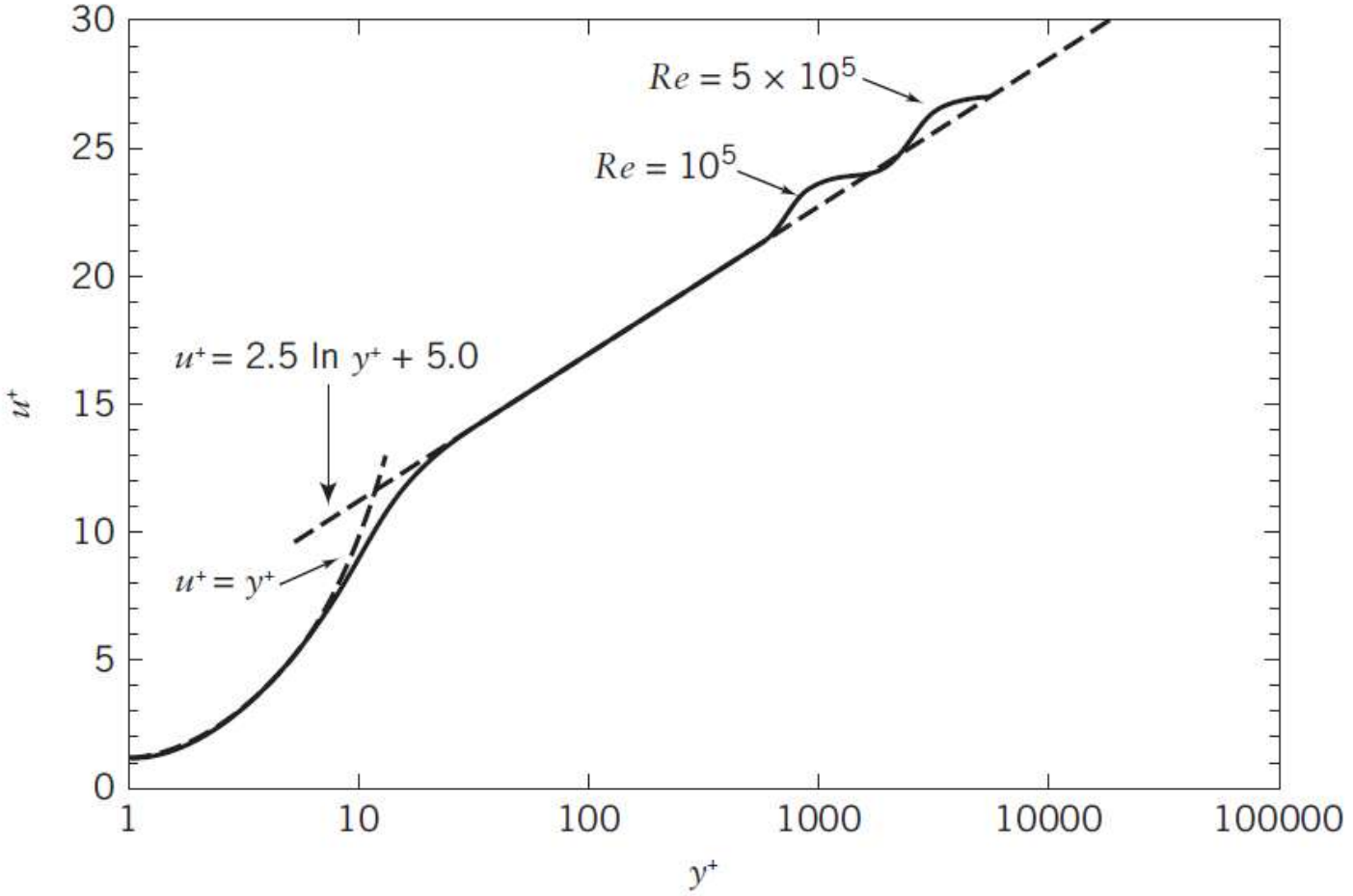
Velocity profiles in fully developed flow

$$\frac{\tau}{\rho} = \nu \frac{d\bar{u}}{dy} - \overline{u'v'}$$

- Prandtl: Turbulent **fluctuations in the mean velocity** are a result of a small element of fluid **moving a small distance** into a **region of higher velocity**.
- The mixing length **increases linearly** with distance from the wall.
- When we are **near the wall**, the total stress is essentially the **wall shear stress**. This region can again be divided into sub-regions.
- **In this sub-region**, when we are **closer to the wall**, the laminar contribution is dominant. This gives us the **viscous sublayer**.
- **In this sub-region**, as we **move away from the wall**, the turbulent contribution is dominant.

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Velocity profiles in fully developed flow



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Revisiting the learning objectives

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